Classifying Constructiveness in an Unfiltered Environment: Automatic Classification of Constructiveness in Social Media Comments Using Large Language Models and Machine Learning

Michelle Wu, Xinlong Chen, Shinji Yamashita, Oscar Wang
Information School
University of Washington
1851 NE Grant Ln
Seattle, Washington 98105

Abstract

In the digital age, platforms like Reddit are key for knowledge exchange and community engagement, though the quality of interactions varies. This study explores automatic classification of constructiveness in social media comments using large language models (LLMs) and machine learning. We developed a system combining manual annotations and models like GPT-4 and BERT to evaluate Reddit comments. Detailed, relevant, and evidence-supported comments are found to be more constructive. By identifying and promoting such comments, our system enhances online discussions, fostering a respectful and informative community. Future work includes expanding the dataset, refining models, and assessing long-term impacts on online discourse.

1. Motivation

In today's digital age, online platforms like Reddit play a significant role in knowledge exchange and community engagement. However, the quality of interactions can vary widely, and not all contributions are constructive or conducive to positive discourse. Constructive feedback is crucial as it fosters learning, promotes understanding, and encourages a respectful exchange of ideas. The importance of enhancing the quality of feedback in online discussions cannot be overstated, as highlighted in studies such as those by White et al. (2012), who demonstrate the educational benefits of constructive feedback in academic settings.

For example, a constructive comment on Reddit might look like this: "I appreciate your perspective on renewable energy. To build on your point, I found a study that suggests solar energy could be more efficient in urban areas due to higher energy demands. What do you think about integrating solar panels on public buildings to maximize efficiency?" This comment not only acknowledges the original post but also provides additional information and invites further discussion, thereby enriching the conversation.

Constructive discussions are relevant to casual online interactions because they help build a positive and engaging community. They encourage users to share valuable insights, support their arguments with evidence, and remain respectful even when disagreeing. The benefits of such discussions include improved mutual understanding, the potential for learning, and the promotion of a more respectful and informed community. Constructive feedback can transform a casual conversation into a meaningful exchange that can have a lasting impact on the participants and the community at large.

Copyright © 2024, Association for the Advancement of Artificial Intelligence (www.aaai.org). All rights reserved.

2. Problem Definition

Our project addresses the challenge of automatically identifying whether a comment on Reddit is constructive. This involves developing a system that can discern the quality and constructiveness of user comments effectively, thereby supporting a more positive and productive online environment. Constructive comments are typically insightful, offer valuable information, and encourage positive interactions. The ability to automatically identify such comments can significantly enhance the user experience by promoting more meaningful engagements and could potentially be used to guide moderation efforts.

Context and Prior Work

Our research breaks new ground by focusing on the constructiveness of comments on social media, diverging from previous studies that often explored comment moderation within the controlled environments of news websites. The dynamics of social media allow for a more unfiltered interaction, providing a broader context for analyzing constructiveness.

We draw upon established principles such as those found in *Constructive Feedback* by Roland and Frances Bee, which advocate for feedback that is specific, relevant, and supportive. These principles form the backbone of our criteria for constructive comments.[BB11]

The landscape of our study is further enriched by leveraging the findings from various platforms like ScienceDirect and ResearchGate, which offer foundational knowledge in text analysis and sentiment assessment. Additionally, we integrate insights from advanced AI research, notably the adaptability and capability of models like GPT-4, to enhance our understanding and analysis of online interactions.

The Constructive Comments Corpus (C3), described in the study on classifying constructive comments, outlines

methodologies for annotating news comments with constructiveness and toxicity, utilizing both classical feature-based and deep learning systems to classify comments. This framework provides a useful reference for our approach to analyze social media comments using natural language processing techniques [Ngu+20].
Furthermore, the exploration of constructive and toxic speech detection for Vietnamese social media comments expands our perspective by demonstrating the application of machine learning in analyzing non-English datasets, emphasizing the global applicability of such research [Zha+20].
These resources collectively guide our methodology, combining manual and automated annotation to strike a balance between human nuance and AI efficiency, aiming for a comprehensive analysis of comment constructiveness across diverse social media platforms.

4. Objective

The primary goal is to build a robust model that can automatically evaluate and classify comments on Reddit according to their constructiveness. By leveraging both manual annotations inspired by educational feedback theories and automated annotations using GPT-4, this project aims to create a tool that not only identifies constructive comments but also enhances the overall quality of online discussions.
A model that can identify and promote constructive comments has numerous benefits for online communities. For instance, in a subreddit dedicated to health and fitness, a constructive comment might offer personal experiences, cite relevant research, and pose thoughtful questions, thereby fostering a more informative and supportive environment. The ability to highlight such comments helps create a space where users feel valued and encouraged to contribute meaningfully.
The broader implications of fostering constructive discussions include the enhancement of online discourse quality, the reduction of toxic interactions, and the creation of a more supportive and educational online environment. By identifying and promoting constructive comments, we can help guide users toward more positive and productive interactions, ultimately contributing to healthier and more engaging online communities.

5. Justification for Choosing Reddit

In our project "Classifying Constructiveness in an Unfiltered Environment," the selection of Reddit as the primary platform for data collection and analysis is strategically advantageous due to its unique features and user base. Here are the reasons that substantiate this choice:

1. **Diverse and Engaged User Base:** Reddit's vast, global community encompasses a wide range of interests, making it a fertile ground for studying interaction patterns and discourse quality across diverse topics and demographics.
2. **Unfiltered and Organic Interactions:** The platform's relatively relaxed moderation policies allow for naturalistic user interactions, essential for studying the spectrum of constructiveness in online communication.
3. **Subreddit Structure:**Organized into subreddits with specific topical focuses, Reddit facilitates targeted data collection, enabling detailed analysis within well-defined thematic boundaries. For our research, we have selected subreddits representing diverse interests: Politics, Entertainment, Science, Sports, and Technology. This selection allows for a comprehensive analysis across different public discourses, crucial for understanding the nuances of online interaction and comment constructiveness. The utility of Reddit as a data collection source, particularly through platforms like Pushshift, is well-documented, providing a robust framework for academic research that necessitates extensive social media datasets. Pushshift, in particular, enhances the accessibility and manageability of Reddit data, which is pivotal for studies requiring large-scale data analysis across various topics.[Bau+20]
4. **Availability of Metadata:** Reddit provides rich metadata, including upvotes, downvotes, and reply chains, which are invaluable for analyzing user engagement and perceptions of content constructiveness.
5. **API Access for Research:** The availability of a comprehensive API enables efficient data extraction, crucial for applying machine learning techniques on a large scale to classify the constructiveness of comments.
6. **Relevance to Academic and Practical Applications:** Studying constructiveness on Reddit not only advances academic knowledge of online communications but also has practical implications for enhancing discourse quality on social media platforms.

By leveraging Reddit, the research captures a broad spectrum of unmoderated, genuine user interactions, providing insights that are directly applicable to enhancing constructive communication in online communities.

Proposed Method

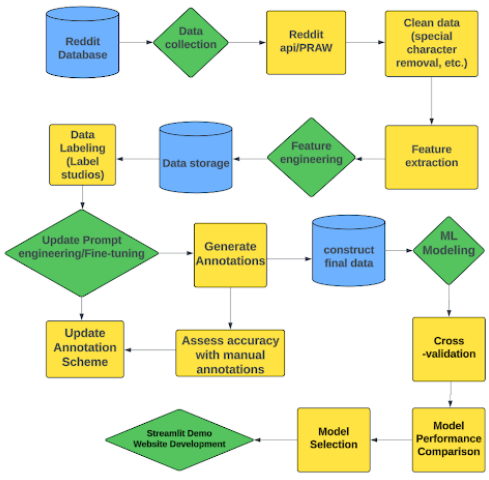


Figure 1: Proposed Method Workflow

Data Collection

The data was collected using the Reddit API and the Python Reddit API Wrapper (PRAW). The script was configured to pull comments from multiple subreddits, focusing on threads with high engagement to ensure a diverse and representative sample of discussions. Here are the key steps in data collection:

Dataset Size and Date Range

The dataset consists of 3,360 comments collected from Reddit, spanning from March 2023 to March 2024. This time frame was chosen to capture a wide range of recent discussions and ensure that the data reflects current trends and topics.

Selection Criteria

- **High Engagement Threads:** Comments were extracted from threads with high engagement, defined by the number of upvotes and the presence of replies. This criterion was used to capture active and dynamic discussions, which are more likely to contain a variety of opinions and levels of constructiveness.
- **Multiple Subreddits:** Data was collected from various subreddits to cover a broad spectrum of topics. The selected subreddits included those focused on Politics, Entertainment, Science, Sports, and Technology, ensuring diversity in the types of discussions analyzed.

Advantages and Limitations

Advantages:

- **Diverse Topics:** By selecting subreddits from different domains, we ensured that the dataset encompasses a wide range of discussions, enhancing the generalizability of our findings.
- **Active Discussions:** Focusing on high engagement threads helps capture comments that are more likely to be detailed and interactive, which are useful for analyzing constructiveness.

Limitations:

- **Bias Towards Popular Discussions:** High engagement threads might introduce biases, as they tend to reflect more popular or controversial topics. This could skew the dataset towards comments that align with majority opinions or popular sentiments, potentially overlooking less visible but equally constructive contributions.
- **Sample Representativeness:** Although the dataset covers a broad range of topics, it might not fully represent the entire spectrum of discussions on Reddit. The selection of only five specific subreddit themes could further introduce bias by not including other diverse or niche communities.
- **Limited Data Handling Capacity:** Given the 10-week timeframe for this project, it was not feasible to handle a larger dataset. While the current dataset size provides a solid foundation for analysis, future work could benefit from scaling up the data collection to include more comments and a wider variety of subreddits.

To mitigate these limitations, future iterations could include random sampling of comments from a wider range of engagement levels and subreddits, as well as expanding the thematic categories to ensure a more balanced and comprehensive dataset. Additionally, exploring methods to reduce potential biases introduced by focusing on high engagement threads will be essential for enhancing the representativeness and reliability of the findings.

Data Collection

- **API Initialization:** Using PRAW, the script initialized a connection to Reddit with appropriate credentials stored in an environment file for security.
- **Comment Extraction:** Comments were extracted based on specific criteria, including the number of upvotes and the presence of replies to ensure that the data represented active conversation threads. This approach aims to capture a range of opinions and engagement levels, but it also introduces potential biases.

Selection of Subreddits For our research, we have selected subreddits representing diverse interests: Politics, Entertainment, Science, Sports, and Technology. These categories were chosen to provide a broad spectrum of topics and discussions. However, it's important to note that this selection is limited to these five themes, which may not fully encompass the entire range of topics and interaction styles present on Reddit.

Potential Biases in Data Collection Focusing on highly engaged threads may skew the dataset towards more popular or controversial topics, which could affect the generalizability of the model to less active discussions. Additionally, the reliance on upvotes and replies might favor comments that align with the majority opinion or popular sentiment, potentially overlooking less visible but equally constructive contributions. The choice of only five specific subreddit themes could further introduce bias by not including other diverse or niche communities.
To mitigate these biases, future iterations could include random sampling of comments from a wider range of engagement levels and subreddits, as well as expanding the thematic categories to ensure a more balanced and comprehensive dataset.

Data Preprocessing

The raw data underwent several preprocessing steps to clean and prepare it for analysis:

- **Text Cleaning:** Regular expressions were used to remove URLs, special characters, and formatting artifacts that could skew the analysis. This step ensures that the text data is uniform and free of irrelevant information.
- **Date Parsing:** Comment timestamps were converted from UNIX timestamps to human-readable dates to facilitate time-based analysis. This allows for the examination of temporal patterns in comment constructiveness.

By addressing potential biases in data collection and implementing thorough preprocessing steps, we aim to create a

robust and representative dataset that supports the development of a reliable model for evaluating constructiveness in Reddit comments.

Data Storage

The cleaned and processed data was stored in a CSV file, ensuring that the dataset was easily accessible for further analysis and model training.

Manual Data Annotation

To ensure the integrity and usability of our dataset for training machine learning models, we conducted a detailed manual annotation on a subset of 500 comments selected from the total of 3,360 comments collected from Reddit. This annotation process was guided by three specific criteria inspired by the principles in "Constructive Feedback" by Roland and Frances Bee. Each criterion was carefully chosen to reflect essential aspects of constructive feedback, and annotators were trained to rate each criterion on a scale from 0 to 1, where:

- 0 indicates the criterion is not met at all,
- 0.5 suggests partial presence or ambiguity in the response, and
- 1 signifies that the criterion is fully met.

Criteria and Rating Implications

- **Personal Experience/Opinion**: This criterion evaluates the inclusion of personal insights or experiences in the comment. A rating of:
  - **0** would be assigned if the comment lacks personal insight or relevance to the discussion.
  - **0.5** could be given if personal insights are mentioned but are not effectively tied to the discussion or only somewhat enhance the conversation.
  - **1** is awarded when the comment integrates personal experience that significantly enriches the discussion, making it more relatable and impactful.
- **Contextual Understanding of Topic**: Comments are assessed based on their reflection of a deep and accurate understanding of the topic discussed. Ratings are applied as:
  - **0** if there is no understanding or irrelevant context is provided.
  - **0.5** if there is some understanding but it's either superficial or partially incorrect.
  - **1** when the comment demonstrates a thorough and accurate understanding of the topic, enhancing the dialogue's relevance and depth.
- **Textual Evidence**: This involves evaluating comments for the presence of supporting evidence or references that bolster the discussion points. The ratings are:
  - **0** if no evidence is provided or if the references are irrelevant.
  - **0.5** if the evidence is somewhat relevant but weak or indirectly supports the discussion.
  - **1** when strong, direct evidence or references are provided, significantly enhancing the comment's credibility and persuasiveness.

Justification for the Criteria Selection The adoption of these criteria stems from the teachings of Roland and Frances Bee, who stress that constructive feedback should be specific, relevant, and supportive. Their framework emphasizes:

- Specificity: Highlighted by the requirement for textual evidence that grounds feedback in verifiable facts or well-supported claims.
- Relevance: Ensured through the evaluation of contextual understanding, making the feedback applicable and beneficial to the discussion.
- Supportiveness: Achieved by encouraging the inclusion of personal insights, which can foster empathy and a more profound connection among participants.

These principles not only guide the evaluation of feedback but also aim to foster a culture of positive and developmental dialogue, enhancing the quality of interactions within online communities. By adhering to these guidelines, our manual annotation process seeks to build a dataset that reflects high standards of constructiveness, ultimately supporting the development of an AI system capable of promoting and maintaining a supportive learning environment on platforms like Reddit.

Initial Weight Determination Using Entropy Weight Method and PCA

To derive preliminary weights for the three scores—Personal Experience/Opinion, Contextual Understanding of Topic, and Textual Evidence—we integrated the Entropy Weight Method (EWM) and Principal Component Analysis (PCA) into our methodology. This dual approach leverages the strengths of each method to ensure a robust and objective evaluation of the criteria's relative importance.

Entropy Weight Method (EWM) The Entropy Weight Method quantifies the dispersion of data for each criterion, reflecting its contribution to the overall diversity of information within the dataset. A higher degree of dispersion suggests a greater potential for differentiation among data points, indicating a higher informational value and thus, a higher weight. Conversely, less dispersion points to redundancy and a lower weight. This method's effectiveness in decision-making scenarios has been critiqued and supported by various studies, which highlight both its advantages in minimizing subjective biases and its limitations under certain conditions, such as excessive zero values that can distort the entropy calculation [ZTY20].

Principal Component Analysis (PCA) PCA complements the EWM by identifying the principal components that explain the majority of variance within the dataset, thereby determining the weight each criterion should carry based on its contribution to these components. This method is crucial for capturing the underlying structure of the data, ensuring that the weights are not only statistically valid but

also meaningful in terms of the dataset's intrinsic characteristics.

Rationale for Combining EWM and PCA Combining EWM and PCA addresses the limitations of using either method in isolation. While EWM provides a measure of variability and uniqueness, PCA offers insights into the data structure, ensuring that weights are aligned with the most significant patterns of variation. This integrated approach ensures a balanced consideration of both dispersion and structural importance, leading to more reliable weighting results.

Calculation Process

1. **EWM Calculation**:

$$E_j = -\sum_{i=1}^{n} p_{ij} \log(p_{ij})$$

   where p_{ij} is the normalized value of the j-th criterion for the i-th comment, calculated as the criterion score divided by the total scores of all criteria for that comment.

2. **PCA**:
   - Perform PCA on the normalized scores to identify the axes of maximum variance.
   - Extract the explained variance ratio for each principal component to determine their contribution to data variance.

By applying these methods, we ensure an objective, transparent, and methodologically sound basis for weighting the criteria, which aligns with the principles of constructive feedback emphasized by Roland and Frances Bee.

6. Weights and data annotation

Analysis and Observations from Weight Determination Using EWM and PCA

EWM Analysis Results In our application of the Entropy Weight Method, we calculated the divergence for each criterion to understand the distribution and significance of the information they contribute to the dataset. The divergence values derived from the entropy calculations are presented in Table 1.

Criterion	Divergence
Contextual	0.009013
Evidence	0.137271
Personal/Experience	0.007498

Table 1: Divergence values for each criterion calculated using the Entropy Weight Method.

These results indicate that the 'Evidence' criterion exhibits a significantly higher divergence compared to 'Contextual' and 'Personal/Experience'. This suggests that 'Evidence' encompasses a broader range of information and plays a more crucial role in differentiating the comments within our dataset. The lower divergence values for 'Contextual' and 'Personal/Experience' imply that these criteria, while still important, offer less variability and therefore, contribute less unique information compared to 'Evidence'.
This focused analysis of divergence highlights the varying levels of information richness and differentiation that each criterion brings to our dataset, providing a clear, quantified basis for their relative importance in the context of manual comment evaluation.

PCA Analysis Results

Following the entropy analysis, Principal Component Analysis (PCA) was performed to further delineate the contribution of each criterion based on how they explain the variance in the dataset. The PCA results were visually represented in a figure, demonstrating that the first principal component (PC1) explains approximately 70% of the total variance. This substantial proportion indicates the significance of PC1 in capturing the primary patterns within the data.

Focus on PC1 Given the dominance of PC1 in explaining the variance, our analysis concentrated on the distribution of weights among the three criteria within this component:

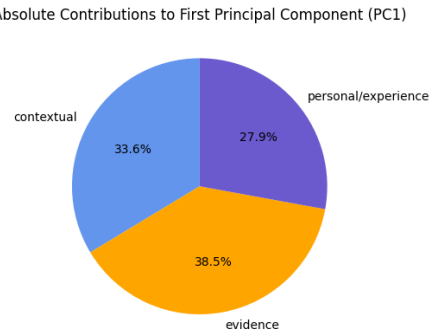


Figure 2: Absolute Contributions to First Principal Component (PC1)

These weights, calculated from the absolute values of the loadings of PC1, normalized to sum to one, suggest that 'Evidence' is the most influential criterion within PC1, contributing to 38.5% of this principal component. 'Contextual' follows with a 33.6% contribution, and 'Personal/Experience' accounts for 27.9%. The higher weight of 'Evidence' aligns with the divergence analysis from the EWM, reinforcing its significant role in the dataset.

Constructive label annotation

Constructiveness Score

After determining these weights by utilizing PCA, we manually labeled 500 out of 3360 comments in our datasets by random using Label Studios. Each comment was scored in three dimensions, as mentioned previously. We took the weighted sum of personal, evidence, and context for each of the four annotators and took the average to be the constructiveness probability defined from 0 to 1 for each comment.

1. **Constructiveness Score Calculation**:

$$\text{construct_prob}_j = \frac{1}{n}\sum_{k=1}^{n}\left(\sum_{i=1}^{3} w_i x_{ijk}\right), \quad \text{for } j = 1, \ldots, m$$

where w_i is the weight of the x_i criterion for the j-th comment by the k-th annotator divided by n annotators. After calculating the construct_prob for each comment, we used a threshold to classify comments into either constructive or non-constructive.

$$y_j = \begin{cases} 1 & \text{if construct_prob}_j \geq 0.5 \\ 0 & \text{otherwise} \end{cases}, \quad \text{for } j = 1, \ldots, m$$

where 1 means that the j-th comment is constructive and 0 means non-constructive.

2. **Annotator agreement matrix**:
Overall, our average agreement rate over all the comments was 80% which we established as our baseline for evaluating the accuracy of LLM models for auto annotation.

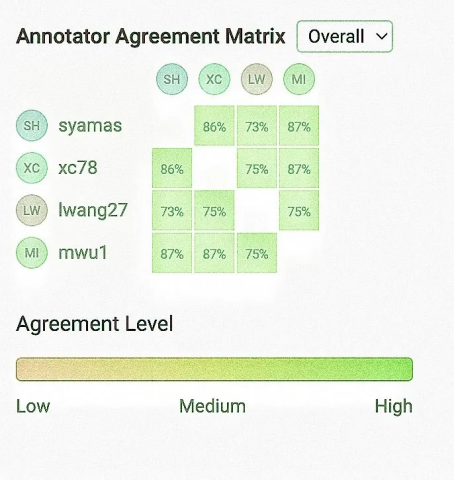


Figure 3: Annotator agreement matrix

Automating annotation using LLM's

After manual labeling, due to our limited time frame, we automated this processing by using LLM's and their reasoning capabilities [Tan+24]. We utilized OpenAI's API to request and regex for extracting the labels from the models. Here is our proposed algorithm which iterates through each comment in our dataframe:
$template_R$: reasoning template instructs the LLM to provide a continous score from 0-1 for each dimension due to average calculation since it creates a score not limited to 0, 0.5 and 1. Otherwise, for mode method, it follows the criteria and rating prompt.

Assessing automated annotations through manually labeled data

We automated our constructive label scoring using two different models: Mistral 7B and GPT3.5-turbo. For Mistral 7B, we utilized Ollama, an application for managing and using open source LLM models. To compute the constructiveness score, we relied on two different computations. The average of the annotators which is mentioned above and the majority for each dimension and calculating the weighted sum afterwards.

Algorithm 1 Procedure to Extract Dimensions of Constructiveness

Require: Original post, Comment
Ensure: Prediction for each dimension: Personal, Evidence, Context

1: Initialize the large language model (LLM) by instruction prompting with $template_R$: reasoning template
2: Input the original post and comment to the LLM
3: Ask LLM to generate scores for three dimensions: Personal, Evidence, Context and a brief justification to faciliate chain-of-thought reasoning.
4: Extract labels for each dimension using regular expressions (Regex):
5: - Use Regex to parse the Personal score from the LLM output
6: - Use Regex to parse the Evidence score from the LLM output
7: - Use Regex to parse the Context score from the LLM output
8: **return** Labels and corresponding justifications for each dimension

Table 2: Performance of Various LLMs

Model and Configuration	Accuracy
Mistral 7B - Few-shot prompting	67%
GPT-3.5-Turbo Base Mode - Few-shot prompting	81.7%
GPT-3.5-Turbo Base Mode - Fine-tuned	80%
GPT-3.5-Turbo Fine-tuned (Baby_3_Average)	84.5%

Note: All tests were conducted with max_tokens=150 and temperature=0.5.

Mistral 7B was tested on 100 manually labeled examples while the rest were tested on 400 examples due to complexity, and slow runtime of the model. Baby_3 and Base-Mode was fine-tuned on 80 examples and 20 for validation. Our goal is to minimize false positives since incorrectly classified constructive comments could be disturbing to users on the social media website. Thus, ROC curve was used to determine the best thresholds for each model. Below is the ROC curve for fine-tuned Baby_3 model.

After a thorough comparison of different LLM models for accuracy, Baby_3 was used to automate all of the data due to its overall accuracy and performance compared to the baseline agreement rate. We did not include the manually labeled data since we wanted to keep it consistent. This was the construct final dataset.

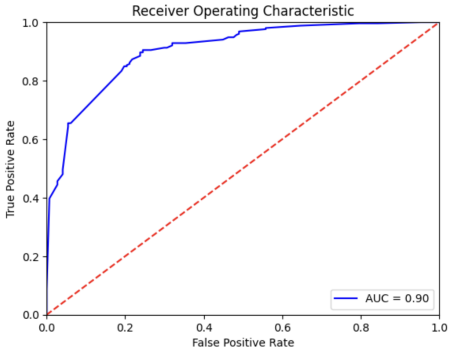


Figure 4: The optimal threshold for baby_3 was 0.6 to decrease the number of false positives

7. Analysis

Feature Correlation Analysis

The correlation matrix provides nuanced insights that highlight the subtleties of online discussions, suggesting strategies to enhance content quality and engagement:

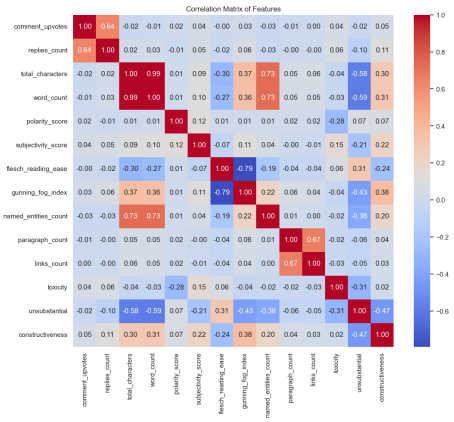


Figure 5: Correlation among different feature

- **Structured Comments and Constructiveness**: The correlation between constructiveness and paragraph count (0.38) underscores that comments with clear organization and structure are perceived as more constructive. This suggests the value of promoting clear communication standards in community guidelines.
- **Credibility through Detail**:
  - **Named Entities and Constructiveness (0.38)**: Comments that include specific references or factual data are seen as more constructive due to their detailed and credible nature.
  - **Encouraging Detailed Contributions**: Platforms could enhance discussions by encouraging users to cite sources or include detailed references.
- **Personal Insights Enhance Discussions**: **Subjectivity Score and Constructiveness (0.24)**: The positive relationship here indicates that personal insights and opinions add perceived value to discussions, suggesting that sharing personal experiences could boost engagement and constructiveness.

These insights collectively advocate for strategies that not only promote clarity and detail in online discussions but also encourage personal contributions, highlighting the importance of both structure and substance in fostering quality interactions and constructive discourse.

Random Forest Model Analysis

We implemented the Random Forest model to classify comments as constructive or not. We split the data into an 80% training set and a 20% testing set to ensure a robust evaluation while training the model with sufficient data.

Feature Importance

Using the intrinsic feature importance method of the Random Forest model, we identified key attributes that contribute to the classification of comments based on constructiveness.

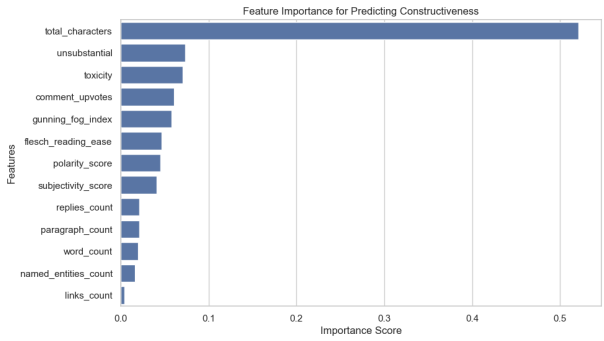


Figure 6: Importance Score Among Features

The following are the top five features, sorted by importance:

- **Total Characters:** Indicative of constructiveness, showing that longer comments tend to be more constructive.
- **Unsubstantial Score:** Reflects the depth of content, with less substantial comments being less constructive.
- **Toxicity:** Less toxic comments are generally more constructive.
- **Comment Upvotes:** There is a correlation between the number of upvotes and constructiveness.
- **Readability Indices:** Includes the Gunning Fog Index and Flesch Reading Ease, indicating that more readable comments are perceived as more constructive.

Feature Engineering (Parts of Speech)

POS frequency analysis

To create new features that were important to improving our prediction, we analysed the the frequency of different parts

of speech between constructive and non constructive comments using Spacy. We took the average of the frequency per comment for each part of speech.

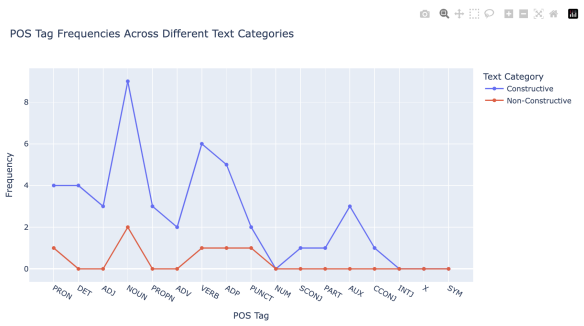


Figure 7: **Parts of Speech analysis**

We determined that there was a noticeable amount of different in the frequency for all parts of speech between non constructive and constructive comments. Constructive comments had a significantly higher amount of *nouns, verbs,* and *prepositions*. Through this, we were able to create more features that had more variance between the classes.

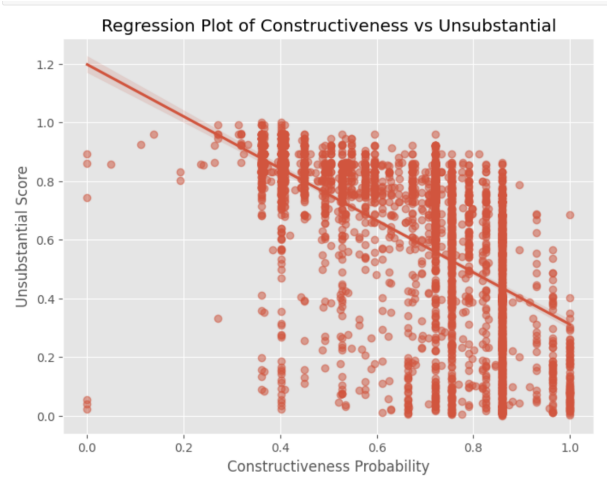


Figure 8: **Unsubstantialness v.s Constructiveness**

This plot shows us an negative correlation between how unsubstantial a comment is versus how constructive a comment is. There is a noticeable trend that the higher probability that a comment is more constructive, the less unsubstantial it is.

Proportion of Constructiveness within Subreddits

The bar chart below provides insight on the proportion of constructiveness scores with the various subreddit categories: entertainment, politics, science, sports, and technology.

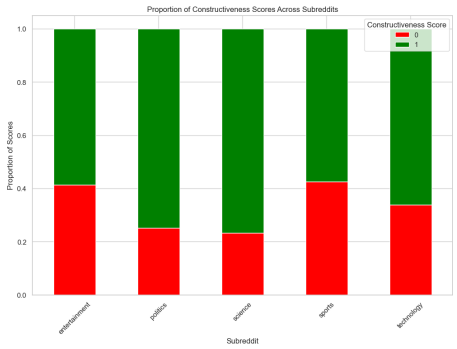


Figure 9: Proportion of Constructiveness Scores

The data presented in Figure 7 above displays a higher proportion of constructive comments in the politics and science subreddits. In science, this could be attributed to the nature of scientific discourse that emphasizes evidence and rational debate. Similarly in politics, despite potential for contentious discussion, it appears that a significant portion of the conversation remains constructive, possibly reflecting a community norm of reasoned debate and information exchange. Overall, the data suggests that the type of subreddit, influenced by its subject matter and the nature of its community, significantly affects the constructiveness of discussions. Subreddits focused on factual and technical content generally foster more constructive interactions, while those revolving around subjective topics face challenges in maintaining the same level of constructive discourse.

Relationships between Criteria and Constructiveness

To further analyze the predefined criteria and influence on constructiveness scores, we created visualizations to quantify and depict how these scores contribute to the overall constructiveness of interactions.
This figure displays three scatter plots, each one illustrating the relationship between constructiveness scores of comments and the three different variable that were used to calculate the resulting score: personal, contextual and evidence scores. A more detailed explanation of each plot is listed below:

- The top plot shows the relationship between personal scores and constructiveness scores. Here, we can see a positive, although not strong, correlation with a moderate spread of data points. The red line indicates a trend that as personal scores increase, the constructiveness scores also tend to increase, which further suggests that comments that contain personal insights or experiences may be more constructive than those that don't.
- The middle plot displays the relationship between contextual scores and constructiveness scores. There is a clearer positive correlation in this plot compared to the one above it, with a steady increase in constructiveness scores with higher contextual scores, indicating a strong dependency. We can infer that comments that are more relevant to the given context or topic are more likely to be rated as more

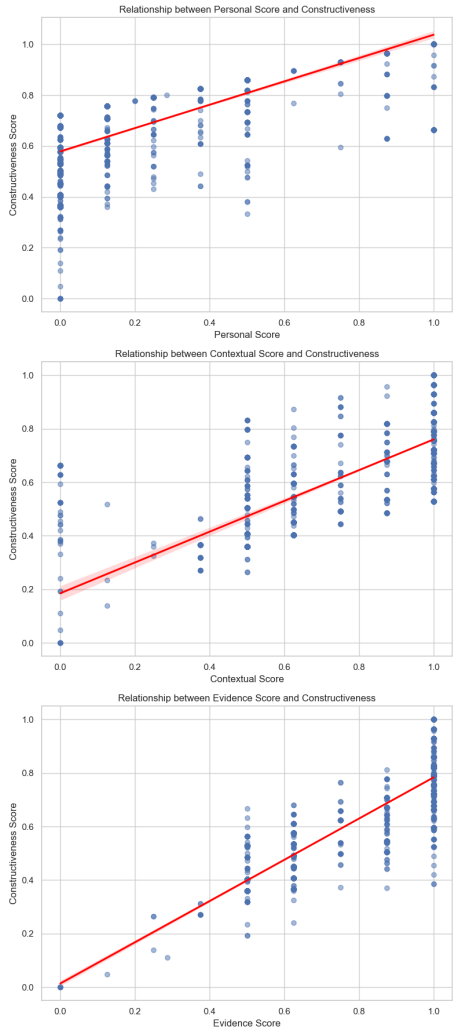


Figure 10: Relationships between Criteria and Constructiveness

constructive, hence suggesting that relevancy to the topic plays a significant role in perceived constructiveness.

- The bottom plot relates evidence scores to constructiveness scores. This plot displays the strongest positive correlation among the three, with constructiveness scores increasing with higher evidence scores as depicted by the steep trend line. The results here indicate that comments that include evidence or factual statements are significantly more likely to be perceived as constructive, which confirms the hypothesis that providing evidence can greatly enhance the constructiveness of a comment.

Comment Length vs. Constructiveness

In addition to the analyses presented thus far, our team was also curious about the relationship between constructiveness and other factors such as comment length. The figure below depicts this relationship and provides some interesting insight on further analysis.

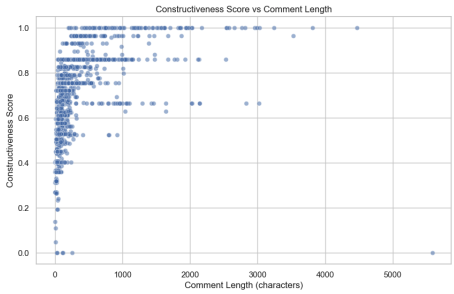


Figure 11: Comment Length vs. Constructiveness

The scatter plot presented in Figure 9 visualizes the relationship between the constructiveness score of comments and their length, measured in characters. From this figure, we can infer the following:

- A significant cluster of data points is observed for comments of relatively short length (up to about 500 characters), where constructiveness scores vary from 0 to 1. As comment length increases beyond 500 characters, the variance in constructiveness scores decreases, with scores generally appearing above 0.4 and many concentrated near 1.0.
- Very few comments have a length exceeding 3000 characters, but those that do maintain high constructiveness scores close to 1.0. This may suggest that longer comments tend to be more thoughtful or well-constructed.
- Similarly, comments that are extremely short (near 0 characters) have constructiveness scores mostly below 0.2, which indicates a lack of constructive content or substantialness.

Overall, there is an apparent positive correlation between comment length and constructiveness score, which suggests that to enhance the constructiveness of discussions, encouraging detailed, well-elaborated comments could be beneficial. Longer comments may be associated with detailed, thoughtful discussions or arguments, thereby receiving higher constructiveness scores, while shorter comments may are less likely to be rated as highly constructive, which aligns with the notion that brevity may often omit detailed reasoning or substantive content.

Incorporating BERT for End-to-End Training

In addition to using feature-based models, we explored an end-to-end training approach using BERT (Bidirectional Encoder Representations from Transformers). This involved fine-tuning BERT models directly on the comments to predict three distinct scores: personal experience score, contextual understanding score, and evidence score. The objective of this approach was to leverage BERT's contextual understanding capabilities to generate precise and nuanced evaluations of comment constructiveness.

Training Methodology

Introduction to the Model and Training Methodology For our project, we employed BERT, specifically the pre-

trained `bert-base-uncased` model, to output three distinct scores for each comment:

- **Personal Experience Score:** Measures the inclusion and relevance of personal insights or experiences.
- **Contextual Understanding Score:** Evaluates the depth and accuracy of the comment's understanding of the topic.
- **Evidence Score:** Assesses the presence and quality of supporting evidence or references.

Each of these scores was produced by separately fine-tuning three BERT models in an end-to-end manner. This allows for a detailed and granular evaluation of the factors contributing to the constructiveness of comments.

Training and Validation Methodology Given the nature of our output, which are continuous scores (logits), traditional classification metrics like accuracy were not suitable. Instead, we used Mean Squared Error (MSE) to validate the performance of our models. To ensure robust and reliable evaluation, we employed 5-fold cross-validation. This method involves splitting the dataset into five subsets, where each subset is used once as the validation set while the remaining subsets are used for training. This technique helps mitigate overfitting and provides a more comprehensive assessment of the model's performance.

Findings and Insights

Through this approach, we observed that the BERT-based models were effective in capturing the nuanced aspects of constructiveness in comments. The continuous scores provided by our models offer a detailed assessment, enabling a more granular understanding of the factors contributing to constructive feedback. The training results demonstrated that BERT could be fine-tuned to accurately predict each of the three scores with relatively low MSE.

Limitations

One limitation of our approach is that the use of logits as output scores means we cannot rely on classification metrics like accuracy. However, by using MSE and k-fold cross-validation, we ensured that our models are well-validated and perform reliably across different subsets of data. Another limitation is the potential for overfitting, which we addressed through careful cross-validation and regularization techniques.

Impact on Constructiveness

The insights gained from our models can significantly enhance the understanding of what makes a comment constructive. By providing separate scores for personal experience, context, and evidence, our model helps identify specific areas where comments excel or fall short. This granular feedback can be used to guide users in crafting more constructive comments, ultimately fostering a more productive and positive online environment.

Conclusion

Our approach demonstrates that fine-tuning BERT for continuous score outputs is a promising method for evaluating and improving the constructiveness of online interactions. The detailed and nuanced scores generated by the models can provide valuable feedback to users, encouraging more meaningful and constructive discussions.

Traditional Machine Learning Methods

Our goal is to predict the probability of a comment being constructive. We decided to remove the scores used to calculate the constructiveness since the models would just relearn the weights without understanding the semantic meaning of the text. Additionally, generating those scores from an annotator model would be too computationally expensive. Our general pipeline for model selection involved a process of comparing SBERT and TF-IDF embeddings and optimizing accuracy and f1-score by selecting the best set of features. For TF-IDF embeddings, we used a new column called cleaned comment which removed spelling errors and lemmatized for further consistency since TF-IDF may be more sensitive.

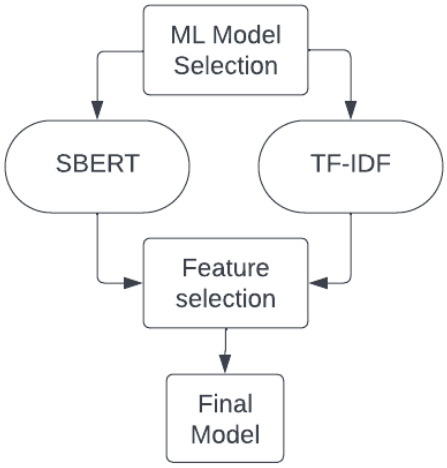


Figure 12: ML pipeline

Embedding and Model explanations

- **SBERT Embeddings:** SBERT embeddings are designed to capture specific nuances and the context of sentences more effectively. They aim to represent semantic meanings derived from both directions of the text, allowing for a deeper understanding of the language.
- **TF-IDF Embeddings:** On the other hand, TF-IDF embeddings focus on representing text based on the frequency of terms within the document and across documents. This method is less capable of capturing deep semantic meanings compared to SBERT, as it primarily quantifies word significance rather than context or nuance.

To address the complexities in our constructiveness classification problem, we utilized four different machine learning models, each chosen for its certain advantages in handling text data.

- **Logistic Regression:** Known for its interpretability and efficiency, making it an excellent baseline for performance comparison.
- **Naive Bayes Classifier:** Known for its capability and speed in handling large, multi-dimensional data. However, assumes all features are indepedent.
- **XGBoost:** Ensemble method that utilizes the power of many small learning models that builds on previous implementations to produce a more accurate model.
- **SVM:** Robust in high dimensional data, leveraging its kernel trick to handle non-linear relationships in text.

Table 3: Model Evaluation Results Ordered by Accuracy

ML Model Type	Embedding	Accuracy (%)
XGBoost	SBERT	83.7
SVM	SBERT	81
Logistic Regression	SBERT	80
XGBoost	TF-IDF	77
Naive Bayes (Gaussian)	SBERT	75
Logistic Regression	TF-IDF	72
SVM	TF-IDF	69.4
Naive Bayes	TF-IDF (Unigram)	68
Baseline	None	67

Model Evaluation results

All models were evaluated without any additional features except for embeddings. One common pattern across all models were that SBERT embeddings performed better than TF-IDF. Among all models, XGBoost paired with Sentence-BERT embeddings performed the best. This is our base model.

Feature Selection

After choosing XGBoost+SBERT embeddings as our base model, we performed KBest feature selection on additional features to help increase the accuracy of our model. We dropped word count since our classifier would incorrectly predict a comment as constructive if it was only long.

XGBoost Final Model

By concatenating the additional features listed above, we were able to gain marginal improvements to our base classifier, which resulted in a 85.1% accuracy. We have a demo that works and due to API issues, specific permission is needed to access it. If you need access, please reach out to us.

Table 4: Features Used in XGBoost Classifier

Feature Type	Description
Unsubstantial	Indicates the lack of substantial content or value in the text by using *Perspective API*.
Subjectivity Score	Measures the amount of personal opinion and subjective information.
Readability Score	Assesses the ease with which the text can be read and understood by using the *Flesch Kinscaid measurement*.
Parts of Speech	Includes frequency or relevance of *nouns, verbs, adjectives, prepositions* and *determiners* in the text.

Deep Learning Approach

Because of the dimensions and size of the text embeddings, machine learning models are not able to capture such relationships effectivitely as they are suitable for more structured data. Instead, we use deep learning approaches as the rise of transformer architectures and encoding only models such as BERT has achieved many numerous benchmark performances on text classification tasks.

- **RoBERTa:** RoBERTa is further fine-tuned on BERT which aims to capture the context of meaning in text from both directions that is trained on a large corpus of text. It has beaten BERT in many benchmark datasets.

Table 5: Comparison of RoBERTa Model Accuracies

Model Type	Accuracy (%)
RoBERTa-Base	82.7
Social-Construct-RoBERTa	84.8

- **RoBERTa-Base:** Fine-tuned on reddit comments only after some epochs.
- **Social-Construct-RoBERTa:** Concatenated the first token last hidden layer with the features selected for the XGBoost model.

The accuracy was lower than expected just by a little bit, however, precision was a bit higher this time in the SoCial-Construct-RoBERTa which is creates a higher standard of learning and bar for constructiveness. 3 epochs were utilized with a learning rate of 1e-5 and batch size of 16 were used. This model is stringent in identifying constructive comments.

Limitations

Model Output Limitations

Our models currently use logits as output scores, limiting the use of traditional classification metrics like accuracy. While Mean Squared Error (MSE) is a valuable measure, it does not fully capture the model's effectiveness in distinguishing

between constructive and non-constructive comments. Future work may involve developing or identifying more suitable evaluation metrics.

Dataset Bias

The current dataset is limited to specific subreddits, which may introduce biases. This selection might not fully represent the diversity of online discourse. Future efforts will focus on broadening the dataset to include a wider variety of topics and interaction styles, ensuring the model's applicability across different contexts.

Knowledge distillation

Our best model's accuracy was close to the fine-tuned GPT-3.5 model's accuracy which may be explained by knowledge distillation. The task-specific model was able to capture the annotator's models knowledge, however, it limited the accuracy of our student model due to the quality of the annotations which could have been assessed more thoroughly given enough time.

Overfitting Concerns

Despite using techniques like k-fold cross-validation, overfitting remains a potential issue. Continuous validation on new, unseen data and implementing advanced regularization techniques will be essential to maintain model robustness and generalizability.

Discussion

Interpretation of Results

Our results indicate that utilizing a hybrid approach of manual and automated methods to classify the constructiveness of comments on Reddit is both effective and scalable. The integration of large language models (LLMs) like GPT-3.5 and GPT-4, along with traditional machine learning techniques such as Random Forest and XGBoost, significantly improved our ability to discern constructive comments.

The findings suggest that comments with higher contextual understanding and evidence scores are perceived as more constructive. This aligns with our initial hypothesis that detailed, relevant, and well-supported comments contribute positively to online discourse. The use of BERT and its ability to generate nuanced scores for different aspects of constructiveness further supports this conclusion.

Connection to Project Goals and Motivation

The primary goal of our project was to develop a robust model that can automatically evaluate and classify comments on Reddit based on their constructiveness, thereby enhancing the quality of online discussions. The results of our study directly address this goal by demonstrating that our hybrid approach can accurately classify constructive comments and provide valuable feedback to users.

The motivation behind this project was to foster more positive and engaging online communities by promoting constructive feedback. The success of our models in identifying and highlighting constructive comments supports the idea that such tools can play a crucial role in improving the quality of online interactions. By encouraging users to provide detailed, relevant, and evidence-based comments, we can create a more respectful and informative online environment.

Implications and Future Directions

The implications of our findings are significant for the development of automated content moderation and feedback systems. By incorporating our models into platforms like Reddit, we can help guide users towards more constructive interactions, ultimately promoting a healthier and more supportive online community.

Future work will involve expanding our dataset to include comments from a wider range of subreddits and other social media platforms to improve the generalizability of our models. Collaborating with Reddit to gain access to more comprehensive data and integrating our tools seamlessly into the platform will also be a key focus. Additionally, conducting longitudinal studies to assess the long-term impact of our interventions will provide valuable insights into the effectiveness of our approach in promoting constructive discourse over time.

Conclusion and Future Work

Our research on the automated classification of constructiveness in social media comments, particularly on the Reddit platform, has provided valuable insights and advancements in the understanding of online discourse dynamics. The midterm results of our study underscore the effectiveness of combining manual and automated methods to enhance the assessment of constructiveness in comments. Specifically, the integration of GPT-4 and other large language models (LLMs) has demonstrated substantial potential in recognizing and categorizing constructive feedback within a diverse and unmoderated environment.

Conclusions

- **Effectiveness of Hybrid Annotation:** The combination of manual annotation influenced by educational feedback theories and automated annotations using LLMs has proven effective. This hybrid approach facilitates a more nuanced understanding and classification of constructiveness, balancing human insight with AI efficiency.
- **Model Performance:** Preliminary testing with models like GPT-3.5 Turbo and the implementation of a Random Forest classifier have shown promising results in identifying constructive comments. These tools have allowed us to handle large datasets effectively, providing a scalable solution for real-time moderation and feedback systems.
- **Importance of Detailed Analysis:** Our analysis has highlighted the significance of various features, such as comment length, readability, and the presence of evidence, in contributing to the constructiveness of comments. These findings can guide the development of more effective content moderation strategies and community guidelines.

Future Work

Exploring Advanced Models and Features

Our next steps will involve exploring more advanced machine learning models and features to enhance the accuracy and robustness of our constructiveness classification system. This includes experimenting with state-of-the-art models beyond BERT, such as GPT-4, RoBERTa, and other transformer-based architectures. We will also delve into additional features, such as sentiment analysis, engagement metrics, and linguistic diversity, to enrich the model's understanding and assessment of constructiveness.

Expanding the Dataset

To improve the model's generalization capabilities, we will expand our dataset to include comments from a broader range of subreddits and other social media platforms. This expansion will help ensure that the model performs well across diverse topics and interaction styles, capturing the full spectrum of online discourse. Collaborating with Reddit and other platforms to gain access to larger and more varied datasets will be a key focus.

Collaboration with Reddit

A significant part of our future work will involve collaborating directly with Reddit. This partnership could provide us with access to more comprehensive data and insights into user behavior, enhancing the effectiveness of our models. Additionally, working with Reddit will allow us to integrate our tools seamlessly into the platform, providing direct benefits to the user community by promoting constructive discourse.

Longitudinal Studies

To assess the long-term impact of our interventions, we will conduct longitudinal studies to monitor changes in the quality of discourse over time. These studies will evaluate how our tools influence user behavior and the overall community environment. By analyzing these trends, we aim to provide evidence-based recommendations for improving online community engagement and maintaining high standards of discourse quality.

References

[BB11] Roland Bee and Frances Bee. *Constructive Feedback*. Universities Press, 2011. ISBN: 8173713243. URL: https://books.google.com/books/about/Constructive_Feedback.html?id=fFLX0vssr7kC.

[Bau+20] Jason Baumgartner et al. "The Pushshift Reddit Dataset". In: *arXiv preprint arXiv:2001.08435* (2020). URL: https://arxiv.org/abs/2001.08435.

[Ngu+20] An Nguyen et al. "Constructive and Toxic Speech Detection for Open-domain Social Media Comments in Vietnamese". In: *arXiv preprint arXiv:2103.10069* (2020). URL: https://arxiv.org/abs/2103.10069.

[Zha+20] Qi Zhang et al. "Classifying Constructive Comments". In: *arXiv preprint arXiv:2004.05476* (2020). URL: https://arxiv.org/abs/2004.05476.

[ZTY20] Yuxin Zhu, Dazuo Tian, and Feng Yan. "Effectiveness of Entropy Weight Method in Decision-Making". In: *Mathematical Problems in Engineering* 2020 (2020), Article ID 3564835. DOI: 10.1155/2020/3564835.

[Tan+24] Zhen Tan et al. "Large Language Models for Data Annotation: A Survey". In: *arXiv preprint arXiv:2402.13446* (2024).